MIL-D-14978A (MU)
30 October 1972
SUPERSEDING
MIL-D-14978 (MU)
28 February 1969

MILITARY SPECIFICATION

DETONATOR, STAB, M55

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of detonator designated as Detonator, Stab, M55.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	-	**Ammunition and Special Weapons, General Specification for**
MIL-I-45607	-	**Inspection Equipment, Acquisition, Maintenance and Disposition of**

STANDARDS

MILITARY

MIL-STD-105	-	**Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)**
MIL-STD-109	-	**Quality Assurance Terms and Definitions**

FSC: 1375

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MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U. S. Army Munitions Command

C8798331	-	Detonator, Stab, M55

PUBLICATIONS

ARMY

ET-8798331	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49	-	Transportation, Parts 0-190

(The Code of Federal Regulations is available from the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402. Orders should specify, "49 CFR 0-190 (latest revision)".)

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Loading.-The loading of the detonator shall comply with all requirements specified on Drawing (Dwg.) 8798331, and with all requirements specified in applicable specifications.

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3.3 Moisture content.

3.3.1 Lead azide.-The moisture content of the lead azide at the loading station at the time of loading the detonators shall not exceed 0.50 percent when determined as specified in 4.4.1.

3.3.2 Primer mixture.-The moisture content of the primer mixture at the loading station at the time of loading the detonator shall not exceed 0.30 percent when determined as specified in 4.4.2.1.

3.4 Functioning.

3.4.1 Sensitivity and energy output.-The detonators shall function satisfactorily as evidenced by a depression in the lead block of a minimum depth of 1/8 inch when tested as specified in 4.4.3.1 or as evidenced by a depression in the steel disc of a minimum depth of .010 when tested as specified in 4.4.3.3.2.

3.4.2 Waterproofness.-The detonators shall be waterproof as evidenced by functioning after water immersion when tested as specified in 4.4.3.3.

3.5 Detonators sticking to packaging.-The detonators shall not stick to the onion skin or spacer in the inner set-up box when inspected as specified in 4.4.4.

3.6 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous; however, a few light scratches not exposing base material may be permitted. All required marking and stamping shall be neat and sharply defined.

3.7 Submission of results of contractor-conducted examinations and tests.-The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract. (See 6.4)

3.8 First article testing.-This specification makes provision for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.

4.2.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.2.2. (See 6.1). The first article sample shall consist of the following:

a. Three hundred and sixty-five (365) detonators

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All parts and completed assemblies submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity, or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process (including heat treatment procedures and equipment), material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.2.2 Inspection to be performed.-All parts and assemblies shall be subjected by the Government to any or all the examinations and tests specified in this specification and any or all requirements of the applicable drawings. In addition, the following tests shall be performed:

4.2.2.1 Sensitivity and energy output (See 3.4.1).-Three hundred fifteen (315) detonator assemblies shall be subjected to this test. If four (4) or more assemblies fail to comply with the applicable requirement, the first article sample shall be rejected. The test shall be performed as specified in 4.4.3.1.

4.2.2.2 Waterproofness (See 3.4.2).-Fifty (50) detonator assemblies shall be subjected to this test. If six (6) or more assemblies fail to comply with the applicable requirement, the first article sample shall be rejected. The test shall be performed as specified in 4.4.3.3.

4.2.3 Rejection.-If any part or assembly fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a part or assembly in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

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4.3 Inspection provisions.

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Metal parts of one interfix lot number from one manufacturer.

b. Primer mixture consisting of batches blended by the same process. Each batch shall be composed of the same ingredients, each of which is to be from one interfix lot from one manufacturer.

c. Lead azide from one interfix lot number from one manufacturer.

d. RDX pellets of one interfix lot number from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (See 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent

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for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 Cup, detonator (See Dwg. 8798332).

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:			
1.	Cup cut, split or cracked	Visual	01001
Major:	None defined		
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.6)	Visual	01002

4.3.2.2 Assembly (See Dwg. 8798331).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Caked explosive on exterior	Visual	02001
2.	Explosive exposed	Visual	02002
Major	AQL 0.65 Percent		
101.	Total length	Gage	02003
102.	Outside diameter	Gage	02004
103.	Crimp radius max.	Gage	02005
104.	Paint on wrong end or paint on both ends or no paint	Visual	02006
Minor	AQL 1.50 Percent		
201.	Distortion	Visual	02007
202.	Disc wrinkled or folded	Visual	02008
203.	Crimp not a full 360 degrees	Visual	02009
204.	Evidence of poor workmanship (See 3.6)	Visual	02010

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4.3.2.3 Box, set-up, inner, after loading and prior to closing slide cover (See Dwg. 9258182).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Felt cushion missing	Visual	03001
2.	Filler missing	Visual	03002
3.	Stiffener missing	Visual	03003
4.	Slide cover assembled reversed	Visual	03004
Major	AQL 0.40 Percent		
101.	Release paper missing or improperly assembled	Visual	03005
Minor	AQL 1.00 Percent		
201.	Marking missing, incomplete or illegible	Visual	03006
202.	Evidence of poor workmanship (See 3.6)	Visual	03007

4.3.2.4 Box, set-up, outer, with inner set-up boxes assembled and prior to assembling top and sealing strip (See Dwg. 9258183).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Red surfaces on inner set-up boxes do not alternate on same side of outer set-up box	Visual	04001
2.	Black surfaces on inner set-up boxes do not alternate on same side of outer set-up box	Visual	04002
Major:	None defined		
Minor	AQL 1.00 Percent		
201.	Marking missing, incomplete or illegible	Visual	04003
202.	Evidence of poor workmanship (See 3.6)	Visual	04004

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4.3.2.5 Box, packing, ammunition for detonator, Stab M55 prior to assembling cover and strapping (See Dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Bags containing set-up boxes damaged	Visual	05001
102.	Cushioning material incomplete or missing from around outer set-up boxes	Visual	05002
Minor	AQL 0.65 Percent		
201.	Evidence of poor workmanship (See 3.6)	Visual	05003

4.3.2.6 Box, packing, ammunition for detonator, Stab M55 (See Dwg. 9258184).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Strapping missing, broken, loose or improperly fastened	Visual-Manual	06001
102.	Box damaged	Visual	06002
103.	Marking missing, broken or illegible	Visual	06003
Minor:	None defined		

4.3.3. Testing.

4.3.3.1 Moisture content of lead azide (See 3.3.1) Major defect, Code No. 07001.-From the quantity of lead azide used to produce a detonator lot, at least a six (6) gram (gm) sample shall be selected for the moisture content determination specified in 4.4.1.

4.3.3.2 Moisture content and composition of primer mixture (See 3.3.3 and Dwg. 7548654) Major defect, Code No. 08001.-From the quantity of lead azide used to produce a detonator lot at least a six (6) gram (gm) sample of primer mixture from each batch shall be selected for the moisture content and composition determinations specified in 4.4.2.

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4.3.3.3 Functioning.

4.3.3.3.1 Sensitivity and energy output Major defect, (See 3.4.1).-Sampling shall be in accordance with MIL-STD-105, Normal Inspection, Table III-A, using a sampling size code letter P and an AQL of 0.40%. The provisions of Para. 8, of MIL-STD-105, regarding reduced or tightened inspection shall apply. Testing shall be as specified in 4.4.3.1 (Code No. 09001).

4.3.3.3.2 After ten (10) successive lots have complied with 4.3.3.3.1 the following may be used, (Code No. 09002):

Sampling shall be in accordance with MIL-STD-105, Normal Inspection Table III A, using a sampling size Code Letter P and an AQL of 0.40 Percent. The provisions of Para. 8 of MIL-STD-105, regarding reduced or tightened inspection shall apply. Testing shall be as specified in 4.4.3.2. If a lot is rejected during this test, the contractor may continue, using this test for the next nine (9) consecutive lots. If all nine (9) consecutive lots meet the acceptance criteria, then this test may remain in effect: however, if any of the nine (9) consecutive lots are rejected, the contractor shall revert back to the test specified in 4.3.3.3.1. In addition, if there is a change in process or source of raw ingredients, the contractor shall revert back to the test specified in 4.3.3.3.1. After ten (10) consecutive lots have again complied with the criteria specified in 4.3.3.3.1, then the test specified herein may be resumed.

4.3.3.3.3 Waterproofness (See 3.4.2) Major defect, Code No. 010001.-The sampling shall be in accordance with MIL-STD-105, Normal Inspection Table II-A, using a sampling size Code Letter H and an AQL of 4.0 Percent. Testing shall be as specified in 4.4.3.3.

4.3.3.4 Inspection of sticking detonators (See 3.5), Major defect, Code No. 011001.-A sample of twenty-five (25) inner set-up boxes from each lot shall be selected for this inspection. No more than one (1) inner set-up box shall be selected from each outer set-up box opened and, in addition, the samples shall be selected a minimum of forty-eight (48) hours after assembly of the inner set-up boxes into the outer set-up boxes. If a total of twenty-one (21) or more detonators from the sample stick to either the release paper or inner set-up box, the lot shall be rejected. The inspection shall be performed as specified in 4.4.4.

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4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 8798331 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3. Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.4 Test Methods and Procedures.-

4.4.1 Determination of moisture content of lead azide (See 4.3.3.1).-A tared dried aluminum dish 2 inches in diameter shall be covered with a sheet of teflon 2 1/2 inches square and sufficiently flat to form a tight cover. An accurately weighed portion of approximately two grams shall be transferred to the moisture dish while both the cover and dish are on the balance pan. The dish and contents shall be transferred to an oven and dried at 95C. plus or minus (±) 2°C. for 2 hours. The cover shall be placed in a sulfuric acid desiccator during the drying period. After drying, the dish shall be quickly transferred to the desiccator and allowed to cool to room temperature. The teflon cover shall be placed on the dish and the assembly weighed. The loss in weight shall be calculated to percentage of moisture in the lead azide as follows:

$$\text{Percentage of moisture} = \frac{A-B}{A} \times 100$$

where:

A = the weight of the specimen before drying.
B = the weight of the specimen after drying.

4.4.2 Test of Primer Mixture.

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4.4.2.1 Moisture content (See 4.3.3.2).-Transfer an accurately weighed portion of approximately one gram of the primer mixture to a tared aluminum weighing dish provided with a teflon cover while both the dish and cover are on the balance pan. Place the dish and contents with the cover removed, in an oven maintained at 60º ± 5°C. for 2 hours. The cover shall be placed in a desiccator during the drying period. After drying, transfer the dish and contents quickly to a desiccator and cool in room temperature. Replace the plastic cover and weigh. Calculate the loss in weight of the dish cover and contents to percent moisture in the sample.

4.4.2.2 Composition (See 4.3.3.2).-The analysis of the composition shall be performed as follows:

4.4.2.2.1 Barium Nitrate.-Transfer an accurately weighed portion of approximately 0.5 grams of the dried sample (dried as described in 4.4.2.1) to a tared medium porosity fritted glass crucible of 30 milliliter (ml.) capacity. Record the weight of crucible and sample as Weight A. Add 3 ml. of distilled water saturated with lead azide at a temperature of 5° ± 2°C. Swirl for 30 seconds, then apply suction. Repeat this procedure two times, making a total of three 3 ml. extractions. Using an eye dropper, transfer to the crucible enough alcohol to thoroughly wet the sample and wash the crucible walls. Aspirate until the odor of alcohol is no longer detectable and dry the crucible in an oven for 30 minutes at 60° ± 5°C. Cool in a desiccator and weigh. Record this weight as Weight B. Reserve the crucible and residue for the normal or basic lead styphnate determination. Correct for the percentage of normal or basic lead styphnate in the water extract as follows: Transfer the filtrate and washings quantitatively to a 100 ml. calibrated volumetric flask, washing the beaker thoroughly with distilled water and dilute to the mark. Using a calibrated pipette, transfer a 10 ml. aliquet of this solution to a second 100 ml. calibrated volumetric flask and dilute to the mark with distilled water. Transfer a portion of this solution to a corex glass spectrophotometric cell (See 6.3) having a width of approximately 1 centimeter (cm) and determine its optical density at a wave length of 410 millimicrons. Use as reference a similar cell containing the same proportion of water saturated with lead azide and alcohol as the sample solution. Since the cell which holds the lead styphnate solution is not identical with the reference cell, correct the optical density obtained as described above for the difference in the amount of light which the two cells scatter and absorb.

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To do this, fill both cells with a solution of water and alcohol, and measure at a wave length of 410 millimicrons, the optical density of the cell which originally contained the normal lead styphnate solution. Calculate the percentage of normal lead styphnate in the filtrate on a dry basis as follows:

Normal lead styphnate found in barium nitrate filtrate, Percent $P_n = \frac{2.94\ (A-B)}{WD}$

Basic lead styphnate found in barium nitrate filtrate, Percent $P_b = 4.48\ \frac{(A-B)}{WD}$

where:

A = Optical density of normal lead styphnate solution.
B = Optical density of water and alcohol solution.
D = Width of corex cell, cm.
W = Weight of dried sample.

Calculate the percentage of barium nitrate in the sample as follows:

$$\text{Barium nitrate, percent} = \frac{100\ (A-B)}{W} - (P_b \text{ or } P_n)$$

where:

W = Weight of the dried sample.
A = Weight of crucible and sample.
B = Weight of the crucible and sample after extractions with water saturated with lead azide.
P_n = Percentage of normal lead styphnate found in barium nitrate filtrate.
P_b = Percentage of basic lead styphnate found in barium nitrate filtrate.

4.4.2.2.2. Basic or normal lead styphnate.-The residue obtained in the determination of barium nitrate shall be extracted with the six-ml portions of ammonium acetate solution at about 25C., (hot ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water, saturated with antimony sulfide, until the washing coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250-ml. calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark.

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Using a calibrated pipette, a 2-ml. aliquot of this solution shall be transferred to a 50-ml. calibrated volumetric flask in the case of basic lead styphnate, or a 2-ml. aliquot of the solution shall be transferred to a 100-ml. calibrated volumetric flask in the case of normal lead styphnate, and diluted to the mark with distilled water. A portion of this solution shall be transferred to a corex glass spectophotometric cell having a width of approximately one centimeter (cm.) and its optical density determined at a wave length of 410 millimicrons. As a reference, a similar cell containing ammonium acetate solution of the same concentration as the solution of the specimen shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the optical density obtained shall be corrected as described above for the difference in the amount of light which the two cells scatter and absorb. This shall be done by filling both cells with ammonium acetate solution and measuring at a wave length of 410 millimicrons, the optical density of the cell which originally contained the lead styphnate solution. The percentage of normal or basic lead styphnate in the specimen shall be calculated as follows:

$$\text{Percentage basic lead styphnate} = \frac{28\ (A-B)}{WD} \pm P_b$$

$$\text{Percentage normal lead styphnate} = \frac{36.75\ (A-B)}{WD} \pm P_n$$

where:

A = Optical density of normal or basic lead styphnate.
B = Optical density of ammonium acetate solution.
D = Width of corex cell, cm.
W = Weight of dried specimen.
P_b = Percent basic lead styphnate found in barium nitrate filtrate.
P_n = Percent normal lead styphnate found in barium nitrate filtrate.

4.4.2.2.3 Tetracene.-The wet residue obtained in the determination of basic lead styphnate shall be washed three times with alcohol, dried in an oven maintained at 60 + 5 C., for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be transferred to a 125-ml. beaker with a stream of water, 25-ml. of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the

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same crucible, and the residue in the crucible washed three times with boiling water and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60 ± 5 C. for 30 minutes, cooled in a desiccator and weighed. The percentage of tetracene in the specimen shall be calculated as follows:

$$\text{Tetracene, percent} = \frac{(C-D)\ 100}{W}$$

where:

C = Weight of crucible and residue after extraction with saturated ammonium acetate solution.
D = Weight of crucible and residue after treating with hot water.
W = Weight of the dried specimen.

4.4.2.2.4 Antimony sulfide.-The weight of the residue shall be calculated from the determination of tetracene to percentage antimony sulfide in the specimen as follows:

$$\text{Antimony sulfide, percent} = \frac{(D-E)\ 100}{W}$$

where:

D = Weight of crucible and specimen after treating with hot water.
E = Weight of crucible.
W = Weight of dried specimen.

4.4.2.2.5 Lead azide.-The lead azide content in the specimen shall be determined by subtracting from 100 the combined percentage of barium nitrate, basic or normal lead styphnate, tetracene and antimony sulfide.

4.4.3 Functioning.

4.4.3.1 Sensitivity and enery output.-The detonator assembly shall be mounted in the test fixture as shown on the inspection equipment list. A weight of 0.25 ± 0.01 ounce shall be dropped from a height of three inches on the firing pin. Satisfactory functioning shall be evidenced by the size of depression in the lead block, when measured, using equipment as shown on the Equipment List.

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4.4.3.2 Sensitivity and energy output.-The detonator assembly shall be mounted in the test fixture as shown on the inspection equipment list. A weight of 0.25 ± 0.01 ounce shall be dropped from a height of three inches on the firing pin. Satisfactory functioning shall be evidenced by the size of depression in the steel disc when measured using equipment as shown on the Equipment List. The steel disc used in this test shall be of the hardness specified on the steel disc drawing.

4.4.3.3 Waterproofness.-The detonators shall be immersed in water to a depth of not less than two inches but not more than three inches for a period of 48 hours, minimum, Both the water and the detonators shall be within the temperature range of 60° to 80° Fahrenheit (F) throughout the test. Within one hour after removal from the water, the detonators shall be tested according to the test procedure for functioning (See 4.4.3.1 or 4.4.3.2).

4.4.4 Inspection for sticking detonators.-Visual inspection shall be performed to determine the defect "detonator sticking to onion skin". A detonator "transfer block" shall then be used to determine if any detonators are sticking to the inner set-up box. The "transfer block" shall be placed against the spacer in the inner set-up box and the box in turn inverted. Visual inspection shall then be performed to determine the defect "detonator sticking to inner set-up box". Any detonators remaining in the box because of being cocked shall not be considered defects. Equipment for this inspection shall be satisfactory to the Contracting Officer.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level C - Minimum Military Package.

5.1.1.1 The detonators shall be preserved and packaged in accordance with Dwg. 9258182.

5.2 Packing.

5.2.1 Level C - Minimum Military Pack.

5.2.1.1 The detonators shall be packed in accordance with Dwg. 9258183.

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5.3 Marking.-Marking shall be in accordance with Dwg. 9258182, Dwg. 9258183, Dwg. 9258184 and Code of Federal Regulations, Title 49, Parts 0-190.

5.4 Shipping.-When components from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of first article samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs submitted by the contractor (See 6.3.5).

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Equipment lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part Number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor, may, however, in connection with non-mandatory designs, with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections must be approved by the Government prior to fabrication or procurement of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category A, Form 3. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

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6.3.5 Submission of contractor design.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-QA-A-S.

6.5 Check test for possible deterioration.-The following test is designed to prevent detonators that have deteriorated from being assembled into major ammunition items. The test shall be performed on detonators from each lot which have been stored for more than two years, or under adverse conditions for any period of time. If the specification for the ammunition item into which the detonators are assembled does not require that the check test be performed, the contracting officer should insert in the contract a provision for the performance of the check test. The contracting officer will arrange for the contractor to be reimbursed for expense incurred in his performance of the check test for possible deterioration.

6.5.1 Functioning.-The sampling plan for the test shall be in accordance with Standard MIL-STD-105, using Code Letter N, single sample, and an AQL of 0.65 percent. The test shall be performed as specified in 4.4.3.1 using equipment in accordance with 4.3.4.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1375-A013

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 65 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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Department of the Army
Headquarters, Picatinny Arsenal
Dover, New Jersey 07801

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Dover, New Jersey 07801

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